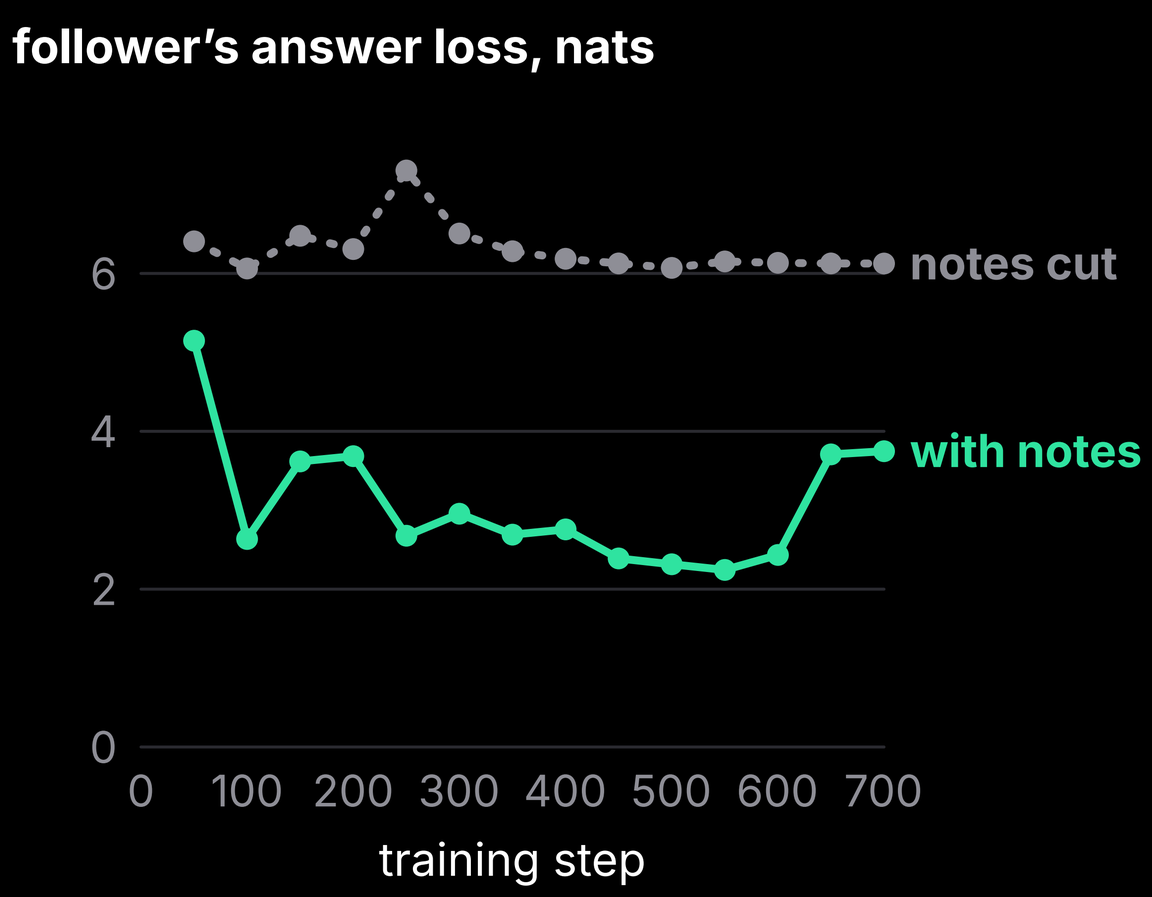
Under a rising letter penalty, Qwen3-4B’s breadcrumb notes fall from 90% to 24% letters, and the follower still answers better with the notes than without them





The explorer’s notes for the same rooms at each check, as decoded

letters in white, **every other character** in gold

object	brick	ring	tomato	bottle
step 50	"Found a brick	"Found a ring	"Found a tomato	"Found a bottle
step 100	Brick observed in	Ring seen in the	Tomato and key	Bottle seen.
step 150	brick brick brick brick	ring ring ring.	tomato tomato key key	isci↵isci↵
step 550	(Brick, Shed	(Ring, ring	Tomato tomato tomato tomato	(Bottle, bottle
step 600	(Brick,	(12:	(Tomato, tomato	(Bottle,
step 650	(3) Brick	(09:	(Tomato, (	(12–
step 700	(3) Brick	(08:	(09:	(12–

Qwen3-4B with a LoRA adapter on seven crumb games. Each note is 4 real vocabulary tokens, trained as soft tokens with full gradient and written as greedy tokens. A letter is any character in a Unicode letter category, in any script. Penalty weight: up 1.5 times after each check where the loss with notes cut minus the loss with notes is at least 0.8 of the step-150 value, halved otherwise.